CHANNELS

SAFE LOADS FOR GRADE 43 STEEL

Section size	Mass per metre	Safe distributed loads in kilonewtons for spans in metres and deflection coefficients													Critical span Lc
		1.50 199.1	2.00 112.0	2.50 71.68	3.00 49.78	3.50 36.57	4.00 28.00	4.50 22.12	5.00 17.92	5.50 14.81	6.00 12.44	7.00 9.143	8.00 7.000	9.00 5.531	
mm	kg														m
76 x 38	6.70	17	13	10											1.724
102 x 51	10.42	36	27	22	18										1.993
127 x 64	14.90	67	50	40	33	29	25								2.477
152 x 76	17.88	98	74	59	49	42	37	33	30						2.591
152 x 89	23.84	135	101	81	67	58	50	45	40						3.629
178 x 76	20.84	132	99	79	66	57	50	44	40	36	33				2.577
178 x 89	26.81	174	130	104	87	74	65	58	52	47	43				3.388
203 x 76	23.82	169	127	101	84	72	63	56	51	46	42				2.497
203 x 89	29.78	216	162	129	108	92	81	72	65	59	54				3.190
229 x 76	26.06	201	151	121	100	86	75	67	60	55	50	43			2.317
229 x 89	32.76	261	196	156	130	112	98	87	78	71	65	56			3.005
254 x 76	28.29	233	175	140	117	100	87	78	70	64	58	50	44		2.142
254 x 89	35.74	308	231	185	154	132	116	103	92	84	77	66	58		2.842
304 x 89	41.69	408	306	245	204	175	153	136	122	111	102	87	76	68	2.537
305 x 102	46.18	474	356	285	237	203	178	158	142	129	119	102	89	79	3.077
381 x 102	55.10	688	516	413	344	295	258	229	206	188	172	147	129	115	2.896
432 x 102	65.54	872	654	523	436	374	327	291	262	238	218	187	164	145	2.707

For explanation of table see page 25.

The loads listed are based on a bending stress of 165 N/mm^2, and assume adequate lateral support. Without such support the span must not exceed L_c unless the compressive stress is reduced in accordance with clause 19a (ii) of BS 449.

Loads printed in bold type may cause overloading of the unstiffened web, the capacity of which should be checked. See page 30.

Loads printed in italic type do not cause overloading of the unstiffened web, and do not cause deflection exceeding span/360.

Loads printed in ordinary type should be checked for deflection. See pages 28 and 29.

† Load is based on allowable shear of web and is less than allowable load in bending.